

Harshana Lakshara Fernando

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Summary

aBackend-focused Computer Science Engineering undergraduate with a rigorous foundation in **System Design**, **Distributed Systems**, and **Cloud-Native Engineering**. Proven experience architecting high-performance, fault-tolerant applications using **Microservices**, **Serverless Architectures**, and **Event-Driven Design** with **Java (Spring Boot)**, **Go**, **AWS Lambda**, and **Kafka**. Validated through the delivery of high-throughput systems handling massive load, seeking to solve complex **concurrency** and **low-latency** challenges within high-velocity engineering teams.

Education

B.Sc. (Hons) in Engineering – Computer Science and Engineering
University of Moratuwa, Sri Lanka

2022 – 2026

G.C.E. Advanced Level (Physical Science Stream)
St. John's College, Panadura, Sri Lanka

2019

Results: 3 A's (Combined Mathematics, Physics, Chemistry)

District Rank: 17

Experience

Software Engineer Intern

Resear Mar 2026 – Present

Empathic Computing Laboratory, University of Auckland (Top 100 QS, Remote)

- **KōreroCare: Secure Mobile–Wearable Digital Phenotyping Infrastructure**

- Designed a research-grade digital phenotyping infrastructure for longitudinal behavioral and physiological data collection across Android, WearOS, and backend services.
- Engineered reliable passive sensing workflows using Foreground Services and WorkManager to sustain high-uptime data capture under Android OS constraints.
- Built resilient synchronization, local buffering, and retry mechanisms to reduce data loss and improve consistency across mobile and wearable data streams.
- Contributed to backend architecture using Go, with service-oriented design for scalable ingestion and secure processing of high-volume time-series data.
- Integrated Keycloak for authentication and identity management, and introduced Traefik as an API gateway for centralized routing in a containerized environment.
- Explored Open Wearables integration to support interoperable ingestion and normalization of wearable data across heterogeneous device ecosystems.

KEYWORDS: Kotlin, Android, WearOS, Go, Keycloak, Traefik, Open Wearables, FastAPI, Docker, WorkManager **Software Engineer Intern**

Dec 2024 – May 2025

Information Systems Associates (Pvt) Ltd — Aviation Technology Company

- **AeroOrder — Revamped Order Management Microservice (Java 17, Spring Boot, Grandle)**

- Engineered a **distributed, high-throughput** unique ID generator for Passenger Name Records (PNRs) using a **Kafka**-backed pool model, ensuring uniqueness and security across scaled instances while adhering to aviation domain standards.
- Optimized the PNR generation service to achieve **25+ PNRs/sec throughput** per node with **<100ms latency**, validating performance and resilience through extensive load, stress, and spike testing with **Apache JMeter**.
- Resolved critical **production** issues, including **Kafka** consumer lag and offset resetting, by implementing robust shutdown hooks and enhanced retry mechanisms to improve fault tolerance and system recoverability.
- Led **Research and Development** for **asynchronous load testing**, establishing a monitoring stack with **Prometheus** and **Grafana** to collect and visualize key performance metrics like queue time and latency.
- Investigated and refactored inefficient **Kafka Streams** processing logic, building a proof-of-concept to validate a more efficient **Producer-Consumer** model that significantly reduced consumer lag and improved system stability.

- Modernized the logging infrastructure by implementing structured **JSON logging**, which removed legacy dependencies and improved log parsing efficiency across all microservices.
- **AeroMart — Legacy Monolithic Passenger Reservation System (Java 8 ,Ant)**
 - Architected and implemented a hybrid integration strategy by embedding a **gRPC** client and **Kafka** consumer into a **legacy Java 8** monolith (AeroMart), enabling seamless, backward-compatible communication with the new **AeroOrder microservices** platform.
 - Developed dynamic, database-driven **feature toggles** to control PNR routing logic, minimizing risk during the phased **production rollout** by allowing for granular control and immediate fallback.
- **Other**
 - Authored comprehensive **design documents**, **sequence diagrams**, and **integration plans** on Confluence; facilitated knowledge transfer sessions to ensure seamless handover to the core development team.
 - Actively contributed to the **agile development lifecycle**, participating in all ceremonies, performing **code reviews**, and collaborating with cross-functional teams to deliver **production-ready** features using **Jira** and **Bitbucket**.

KEYWORDS: Java 18, Java 8, **Spring Boot**, **Spring Security**, Ant, **gRPC**, **Kafka**, React, Protobuf, REST API, Couchbase, Oracle DB, Oracle SQL, PostgreSQL, **JMeter**, Gatling, Ant Design, **Docker**, Docker Compose, Gradle, **Prometheus**, **Grafana**, Kibana, Confluence, Jira, Bitbucket, Microservices, **Kafka Streams**, JSON

Projects

See more projects on my portfolio [↗](#)

1. Employee Management System (EMS) (2025 | Individual) Architecture | Github

Designed and built a comprehensive, cloud-native microservices application for employee and department management, featuring a full-stack architecture with a React/TypeScript frontend and Spring Boot backend.

- Architected a distributed system using **Spring Cloud** for service discovery, centralized configuration, and an API gateway, with **Resilience4j** for robust fault tolerance.
- Enabled dynamic configuration management using **Spring Cloud Bus** with **RabbitMQ** and established a robust observability stack (Zipkin, ELK, Prometheus/Grafana).
- Designed and managed a CI/CD pipeline using **GitHub Actions**, with **ArgoCD** for GitOps-based deployments to the Kubernetes cluster.

Keywords: Java 17, Spring Boot, Spring Cloud, React, TypeScript, Resilience4j, RabbitMQ, Docker, Kubernetes, GitHub Actions, ArgoCD, Prometheus, Grafana

2. GoTogether (2024 | Group) Architecture | Live | Github

Developed and deployed a full-stack travel assistant platform with a **Next.js** frontend (SSR/CSR, secure auth) and microservices in **Spring Boot** and **Go**, enabling AI itinerary generation via **Gemini API**.

- Implemented a **Spring Boot** social service with **Keycloak** integration for identity management and a dedicated **Go** service using **gRPC** for high-performance inter-service communication.
- Configured production-grade security with **NGINX**, **Certbot**, and **DuckDNS**, and orchestrated deployment on **Oracle Cloud Free Tier** using custom scripts and Docker optimizations.
- Built a dedicated **Auth Service using Keycloak**, managing OAuth2/OIDC flows, token issuance, and automated realm/client boot-time import scripts.

Keywords: Next.js, Go, Spring Boot, Keycloak, PostgreSQL, NGINX, Docker, Gemini API, Oracle Cloud, gRPC

3. Serverless Auction Platform (2025 | Individual) Architecture | Github

A capstone project built using AWS, following a microservices architecture with an event-driven, serverless (FaaS) design.

- Architected an event-driven system using **AWS Lambda**, **API Gateway**, and **EventBridge**, with **SQS** and **SNS** for asynchronous communication.
- Designed a high-performance **DynamoDB** data model with GSIs and implemented a custom **Lambda Authorizer** for JWT-based security.
- Utilized **LocalStack** and **Docker** for local cloud emulation and integrated **Middy** middleware for robust error handling.

Keywords: Node.js, Serverless Framework, AWS Lambda, DynamoDB, SQS, SNS, SES, JWT, Middy, LocalStack, Docker

Other Relevant Projects

4. Automated AI Shorts Video Creator (2025 | Individual)

An open-source project to automate the creation of short-form videos using a powerful combination of AI tools and workflow automation.

- Orchestrated a video creation pipeline using **n8n** as the central workflow engine.

- Integrated various AI services for script generation, text-to-speech, and content preparation, including **Together AI**, **OpenRouter**, and **ElevenLabs**.
- Containerized the entire stack using **Docker Compose** for one-click local deployment.

Keywords: n8n, Docker, AI, Workflow Automation, Video Generation

5. Food Delivery Application (2024 | Group)

Github 

Built a scalable, event-driven food delivery platform inspired by systems like Uber Eats, using a modern microservices architecture. As part of the core team, I focused on designing and developing the Restaurant Service.

- Developed the Restaurant Module with secure sign-up/login using **Spring Security** and **JWT** (with extended claims).
- Built CRUD APIs for menu and item management; integrated **Redis caching** to improve performance and reduce database load, with appropriate cache invalidation strategies.
- Enabled order sync via **Kafka**: when an order is placed, an event is published to the 'order_created' topic, which the restaurant service consumes to update order details.

Keywords: Java, Spring Boot, gRPC, Kafka, Redis, PostgreSQL, WebSocket, Docker, React.js, Ant Design

6. Go gRPC GraphQL – Multi-Tenant Microservices (2025 | Individual)



Architected a **polyglot, multi-tenant microservices** platform with a focus on scalability, fault isolation, and unified data access.

- Developed backend services in **Go** using **gRPC** and **Protobuf** for efficient inter-service communication.
- Implemented a **GraphQL API Gateway** to provide a single, unified endpoint for client applications.
- Designed multi-tenant data storage with **PostgreSQL**, **Elasticsearch**, and optional **Cassandra** for high-volume workloads.

Keywords: Go, gRPC, GraphQL, Protobuf, PostgreSQL, Elasticsearch, Cassandra, Docker Compose

7. RPAL Interpreter – Functional Language Compiler (2024 | Group)



Engineered a complete compiler and interpreter for the RPAL (Right-reference Pedagogic Algorithmic Language), built from scratch to parse and evaluate programs according to official language grammar and semantics.

- Developed a Lexical Analyzer using custom Finite State Automata (FSA) and regex-based token identification.
- Built an Abstract Syntax Tree (AST) and Standard Tree (ST) generator.
- Implemented a CSE Machine to evaluate programs using 13 semantic rules of RPAL.
- Built a robust testing framework with **Pytest** covering over 168 test cases across AST, ST, and full output validation.

Keywords: Python, Pytest, GitHub Actions, Makefile, CLI

Professional Training, Coursework & Certificates

See all certificates 

- | | |
|---|---|
| <ul style="list-style-type: none"> • Certified Kubernetes Administrator (CKA) Training
<i>KodeKloud</i> (Mumshad Mannambeth) — Completed 27h curriculum with 50+ hands-on labs. | <p><i>Apr 2025</i>
View certificate</p> |
| <ul style="list-style-type: none"> • AWS Certified Solutions Architect Associate (SAA-C03) Training
<i>Udemy</i> (Stephane Maarek) — Completed 27.5h comprehensive cloud architecture course. | <p><i>Aug 2025</i>
View certificate</p> |
| <ul style="list-style-type: none"> • Apache Kafka Series (v3) – Learn Kafka for Beginners
<i>Udemy</i> (Stephane Maarek) — Full System Design & Setup. | <p><i>Mar 2025</i>
[Part 1] [Part 2] [Part 3]</p> |
| <ul style="list-style-type: none"> • 100 Days of DevOps (Bootcamp)
<i>KodeKloud</i> — Intensive hands-on training covering Linux, Python, Docker, Jenkins, Terraform & Ansible. | <p><i>Nov 2025</i>
View certificate</p> |
| <ul style="list-style-type: none"> • Implementing Software Architecture for Large-Scale Systems
<i>Udemy</i> | <p><i>Apr 2025</i>
View certificate</p> |
| <ul style="list-style-type: none"> • Advanced OAuth Security
<i>Udemy</i> | <p><i>Mar 2024</i>
View certificate</p> |
| <ul style="list-style-type: none"> • Next.js: Build Scalable React Apps with Page & App Routers
<i>Udemy</i> — Instructors: Anton Voroniuk, Denys Kohut, Anton Voroniuk Support | <p><i>Nov 2025</i>
View certificate</p> |

Skills

Core	Java Spring Boot Microservices Distributed Systems System Design REST APIs
Backend	Spring Cloud Spring Security Resilience4j Node.js Express.js Gin
Messaging & APIs	Kafka RabbitMQ gRPC GraphQL WebSockets Protobuf JWT OAuth 2.0
Data	PostgreSQL MySQL Oracle DB Redis Couchbase
Cloud & DevOps	AWS Docker Kubernetes CI/CD (GitHub Actions, Jenkins)
Testing	JMeter Gatling Pytest Unit Testing
Tools	Git Gradle Postman Jira Confluence

References available upon request.